PROGRAM 7: Develop a Domain-Specific Chatbot Using LangChain and a Vector Database

AIM

To develop a domain-specific chatbot that answers questions about a selected subject using LangChain, FAISS and a language model.

Domain: Artificial Intelligence.

ALGORITHM

1. Install LangChain, FAISS and model libraries.
2. Prepare AI-related documents.
3. Generate document embeddings.
4. Create a FAISS vector database.
5. Accept a question from the user.
6. Retrieve relevant documents.
7. Pass the retrieved context to the language model.
8. Generate and display a domain-specific answer.

CODE

Cell 1: Install libraries

```
!pip -q install langchain langchain-community langchain-huggingface faiss-cpu sentence-transformers transformers sentencepiece
```

Cell 2: Python program

```
import torch

from transformers import pipeline

from langchain_core.documents import Document

from langchain_community.vectorstores import FAISS

from langchain_huggingface import HuggingFaceEmbeddings

from langchain_core.prompts import PromptTemplate


# Load embedding model

embeddings = HuggingFaceEmbeddings(

    model_name="sentence-transformers/all-MiniLM-L6-v2"

)
```

```
# AI domain documents
documents = [
    Document(page_content="Artificial intelligence enables machines to perform intelligent tasks."),
    Document(page_content="Machine learning learns patterns from data."),
    Document(page_content="Deep learning uses neural networks with multiple layers."),
    Document(page_content="Natural language processing helps computers understand human language."),
    Document(page_content="Computer vision helps computers analyze images and videos.")
]

# Create vector database
vectorstore = FAISS.from_documents(documents, embeddings)

# Load language model
generator = pipeline(
    "text2text-generation",
    model="google/flan-t5-small",
    device=0 if torch.cuda.is_available() else -1
)

# User question
question = "What is natural language processing?"

# Retrieve relevant documents
retrieved_docs = vectorstore.similarity_search(question, k=2)
context = "\n".join(doc.page_content for doc in retrieved_docs)

# Create prompt
prompt_template = PromptTemplate.from_template(
```

```
    "Answer the question using only the context.\n"
    "Context: {context}\n"
    "Question: {question}\n"
    "Answer:"
)

prompt = prompt_template.format(
    context=context,
    question=question
)

# Generate answer
answer = generator(
    prompt,
    max_new_tokens=100,
    do_sample=False
)[0]["generated_text"]

print("Domain: Artificial Intelligence")
print("Question:", question)
print("\nRetrieved Documents:\n", context)
print("\nChatbot Answer:\n", answer)
```

INPUT

What is natural language processing?

OUTPUT

Domain: Artificial Intelligence

Question: What is natural language processing?

Retrieved Documents:

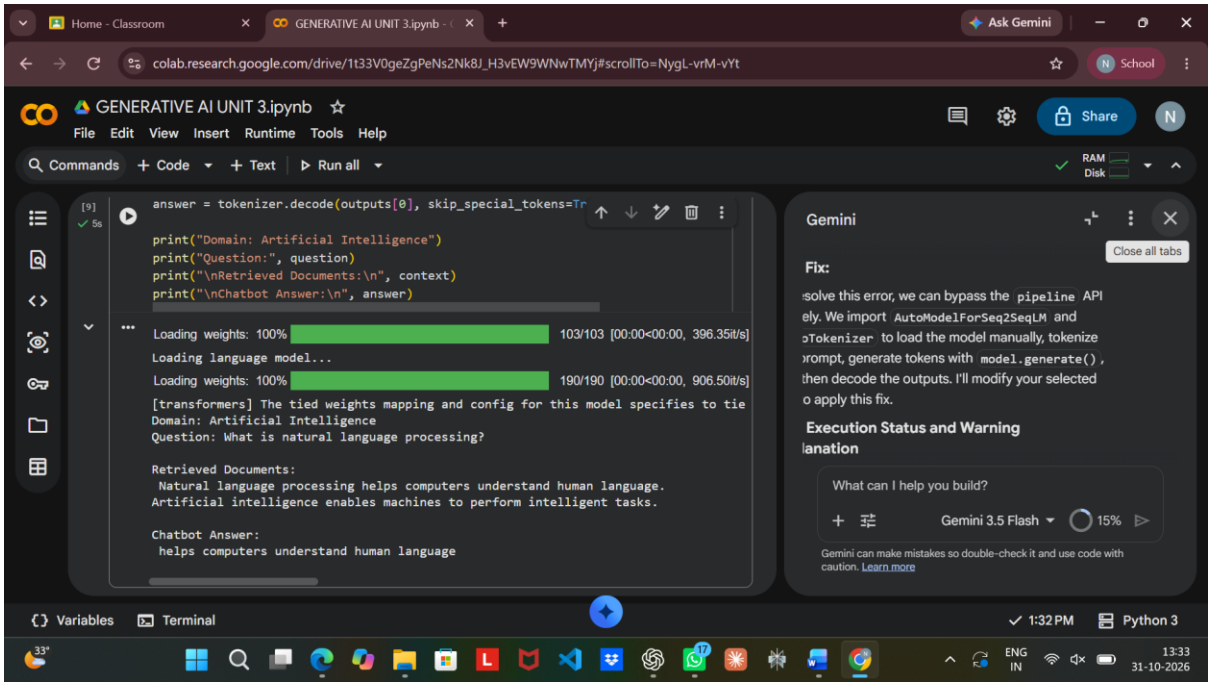
Natural language processing helps computers understand

human language.

Chatbot Answer:

Natural language processing helps computers understand

human language.



RESULT

A domain-specific AI chatbot was developed using LangChain, FAISS, embeddings and a text-generation model.